Failure of finite-jet determination for small-time local controllability

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Abstract

Small-time local controllability asks whether a system can reach a neighborhood of an equilibrium in arbitrarily short time. We give a negative answer to the finite-jet determination question for real-analytic systems with bounded measurable controls. Starting from a previously constructed five-dimensional polynomial system with two controls, we show that adding a positive polynomial term of arbitrarily high order destroys this property. The original system is fixed independently of the prescribed jet order. The proof compares a logarithmic bound for closed trajectories with the positive integral cost of an auxiliary coordinate's excursion.

1 The finite-jet question and the example

Consider real-analytic vector fields $X_0, \dots, X_m$ near $0 \in \mathbb{R}^n$, with $X_0(0) = 0$, and the control-affine system

$$\dot{x} = X_0(x) + \sum_{j=1}^{m} u_j(t) X_j(x), \qquad x(0) = 0, \qquad u(t) \in [-1, 1]^m.$$

Controls are measurable and trajectories are absolutely continuous solutions of the equations almost everywhere. Write $\mathcal{R}_{\le t}(0)$ for the endpoints reachable in a duration at most t. The system is small-time locally controllable at zero if $\mathcal{R}_{\le t}(0)$ contains a Euclidean neighborhood of zero for every sufficiently small $t > 0$.

The finite-jet question asks whether, for each such controllable system, there is an integer N such that every real-analytic tuple with the same Taylor coefficients through order N at zero is again small-time locally controllable with the same control set. This is Conjecture 1.1 in the v7 preprint of Jafarpour [1]. His Theorem 7.1 proves finite-jet stability under a polynomial lower bound on reachable neighborhoods. Jin [2] constructed a polynomial system that is small-time locally controllable without such a lower bound. We show that the same system has no finite jet guaranteeing controllability. The system and the logarithmic estimate used below are from [2]; the additional argument controls the positive cost of a high-order perturbation for every admissible control.

Let e_j denote the jth standard coordinate vector in $\mathbb{R}^5$. The original system is

$$\begin{aligned} &\dot{x}_1 = u, && \dot{x}_2 = x_1, \quad \dot{x}_3 = x_2, \\ &\dot{x}_4 = x_1^2 x_3^2 - \left(\frac{1}{9} + x_5^2\right) x_2^4, && \dot{x}_5 = v, \end{aligned} \tag{1}$$

with measurable $|u|, |v| \le 1$. Denote its drift by X_0 and its controlled fields by $X_1 = e_1$ and $X_2 = e_5$. All three fields are polynomial and $X_0(0) = 0$.

Theorem 1. *The fixed system* (1) *is small-time locally controllable at zero. For every integer* $N \geq 0$*, there is an integer* $k \geq 1$ *with* $2k > N$ *such that the polynomial tuple*

$$Y_0 = X_0 + x_5^{2k} e_4, \qquad Y_1 = X_1, \qquad Y_2 = X_2 \tag{2}$$

is not small-time locally controllable at zero with the same control set $[-1,1]^2$*, although*

$$D^\alpha Y_j(0) = D^\alpha X_j(0) \qquad (0 \leq j \leq 2,\ |\alpha| \leq N).$$

Here α *is a multi-index and* $|\alpha|$ *its total order. This five-dimensional example disproves the universal finite-jet implication for real-analytic systems in dimensions at least three.*

The five-dimensional example suffices for this universal question. No separate conclusion in dimensions three or four is asserted. The perturbation in (2) adds a nonnegative term to the fourth equation. Its integral is positive whenever the fifth coordinate leaves zero, and this contribution will dominate every possible negative fourth displacement in sufficiently small time.

2 Positive perturbations of arbitrarily high order

Theorem 1 of [2] proves the controllability of (1). Its proof constructs trajectories that return the first three and fifth coordinates to zero while producing either sign of fourth displacement, and then steers the other four coordinates, reaching a full neighborhood in a duration at most $5t/8$ for each $t > 0$. We also use the following estimate, stated here with its full hypotheses.

Lemma 2 ([2, Lemma 2])**.** *Let* $T > 0$ *and let* $f \in C^2([0,T])$ *have Lipschitz second derivative,* $\|f'''\|_{L^\infty} \leq 1$*, and*

$$f(0) = f'(0) = f''(0) = f(T) = f'(T) = f''(T) = 0.$$

Set

$$A = \int_0^T f'^4 \,\mathrm{d}s, \qquad E = \int_0^T \left(ff'' + \frac{1}{3} f'^2 \right)^2 \mathrm{d}s.$$

Then

$$E = \int_0^T f^2 f''^2 \,\mathrm{d}s - \frac{1}{9} A. \tag{3}$$

If $A > 0$*, then* $A \leq T^9/16$ *and*

$$E \geq \frac{A}{36 \log^2(T^9/(4A))}. \tag{4}$$

The identity (3) follows by integration by parts. The quantitative lower bound (4) is the logarithmic estimate proved in [2]. We retain the original normalization of f, A, E.

Proof of Theorem 1. The original system's controllability is the cited Theorem 1 of [2]. Fix an integer $k \geq 1$ and consider the perturbed tuple (2). Its first, second, third and fifth equations are unchanged, while

$$\dot{y}_4 = y_1^2 y_3^2 - \left(\frac{1}{9} + y_5^2 \right) y_2^4 + y_5^{2k}.$$

Successive integration gives a unique trajectory on every finite interval for each bounded measurable control. In particular, no existence or finite-time escape restriction enters the argument. We use the usual integral calculus for absolutely continuous functions [3, Appendix 3.A, Example 3.59 and Theorems 3.60, 3.62].

Take any control of duration $T > 0$ whose trajectory ends on the fourth axis, and put $f = y_3$ and $g = y_5$. The unchanged equations give $f' = y_2$, $f'' = y_1$ and $f''' = u$ almost

everywhere. Thus f satisfies every hypothesis of Lemma 2, including the six zero endpoint jets. Also $g(0) = g(T) = 0$ and g is 1-Lipschitz, since $g' = v$ almost everywhere. Define

$$a = \max_{0 \le s \le T} |g(s)| \le T, \qquad P_k = \int_0^T g(s)^{2k} \, \mathrm{d}s.$$

The fourth displacement is exactly

$$y_4(T) = E - \int_0^T g^2 f'^4 \, \mathrm{d}s + P_k, \tag{5}$$

by (3).

If $a = 0$, this displacement is $E \ge 0$. If $A = 0$, continuity gives $f' = 0$, hence $f = 0$, $E = 0$, and $y_4(T) = P_k \ge 0$. If $E \ge a^2 A$, then (5) also gives $y_4(T) \ge 0$. It remains to consider $a > 0$, $A > 0$ and $E < a^2 A$. Since $A \le T^9/16$, the logarithm in (4) is positive. That estimate now gives

$$\frac{1}{36 \log^2(T^9/(4A))} \le \frac{E}{A} < a^2,$$

and therefore

$$A < \frac{T^9}{4} \exp\left(-\frac{1}{6a}\right). \tag{6}$$

The excursion in g also forces a positive lower bound for P_k. Choose $s_* \in [0, T]$ with $|g(s_*)| = a$. The Lipschitz bound and $g(0) = 0$ imply $s_* \ge a$ and

$$|g(s)| \ge a - (s_* - s) \qquad (s_* - a \le s \le s_*).$$

This estimate holds whether $g(s_*)$ is positive or negative. Hence

$$P_k \ge \int_{s_*-a}^{s_*} (a - s_* + s)^{2k} \, \mathrm{d}s = \frac{a^{2k+1}}{2k+1}. \tag{7}$$

Combining (5), (6) and (7), we obtain

$$y_4(T) > \frac{a^{2k+1}}{2k+1} - \frac{T^9 a^2}{4} \exp\left(-\frac{1}{6a}\right). \tag{8}$$

The positive power series for the exponential gives

$$\exp\left(-\frac{1}{6a}\right) \le (2k)! 6^{2k} a^{2k}.$$

Writing $C_k = (2k)!6^{2k}$ and using $a \le T$, the comparison becomes

$$y_4(T) > a^{2k+1} \left(\frac{1}{2k+1} - \frac{C_k T^{10}}{4}\right).$$

Set

$$\tau_k = \min\left\{1, \left(\frac{2}{(2k+1)C_k}\right)^{1/10}\right\} > 0. \tag{9}$$

For $0 < T \le \tau_k$, the expression in parentheses is at least $1/(2(2k+1)) > 0$. Together with the previous cases, this proves that every fourth-axis endpoint in every such duration has nonnegative fourth coordinate.

For each $0 < t \le \tau_k$, the union of endpoints over all durations $0 \le T \le t$ therefore excludes every point $-re_4$ with $r > 0$. Every neighborhood of zero contains such a point, so the perturbed system is not small-time locally controllable.

Finally, given any finite $N \ge 0$, choose $k \ge 1$ with $2k > N$. The only changed field component is x_5^{2k} in the drift. Every partial derivative of this monomial of total order at most N vanishes at zero: a derivative in another coordinate is identically zero, and derivatives only in x_5 retain a positive power of x_5. The controlled fields are unchanged and $Y_0(0) = 0$. The tuple X has been fixed throughout, completing the asserted finite-jet counterexample. □

AI-use statement. OpenAI Codex was used for mathematical development, source checks, drafting and internal manuscript review.

References

[1] S. Jafarpour, *On small-time local controllability*, SIAM Journal on Control and Optimization **58** (2020), 425–446. doi:10.1137/16M1068797. The conjecture and theorem locators refer to the v7 preprint (2019).

[2] K. Jin, *Small-time controllability without polynomial reachable neighborhoods*, research note, October 12, 2026. Versioned public note, Theorem 1, Lemma 2 and Section 3.

[3] J. K. Hunter, *Notes on Partial Differential Equations*, Appendix 3.A, Absolutely continuous functions, Example 3.59 and Theorems 3.60, 3.62, p. 79.